

RTL-to-GDSII Implementation of a Low-Power SoC Using OpenLane

Design, Physical Implementation, and Verification

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Electronics and Communication Engineering

Technology and Tools

OpenLane | SKY130A | sky130_fd_sc_hd

Implementation Scope

Verilog RTL → Synthesis → Physical Design → DRC/LVS → GDSII

Personal VLSI / ASIC Design Study

2026

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1 Introduction

The RTL-to-GDSII flow is the sequence of design and verification steps used to transform a digital hardware description into a physical integrated-circuit layout. The process includes RTL simulation, synthesis, floorplanning, placement, clock-tree synthesis, routing, timing analysis, power estimation, and physical verification before final GDSII generation [1, 2, 3].

This project was undertaken to obtain practical experience with the complete digital physical-design flow using the open-source OpenLane framework. The work follows the design from Verilog RTL through functional simulation, logic synthesis, physical implementation, timing and power analysis, DRC, LVS, and final GDSII generation.

The design implemented in this work is a small system-on-chip named `lowpower_soc`. It integrates a simple CPU, a 256×8 -bit memory, an 8-bit GPIO interface, and a timer. The design also includes a sleep-controlled enable mechanism intended to reduce unnecessary sequential activity when the system is inactive. Enable-based and clock-gating techniques are widely used to reduce unnecessary switching activity in low-power digital systems [10].

The physical implementation was performed using the SKY130 open-source process design kit and the `sky130_fd_sc_hd` standard-cell library [8]. The implementation was evaluated using area, power, timing, and physical verification results. Placement and routing stages were also examined to understand the transformation of the RTL design into its physical representation [4, 5].

Particular attention was given to timing and power because these metrics provide important information about the quality of the implemented design. Static timing analysis was used to evaluate setup and hold behavior [7], while power analysis was used to obtain internal, switching, and leakage power estimates. DRC and LVS were then used to verify the physical implementation.

The final GDSII layout was generated and inspected using KLayout. The completed implementation therefore provides both quantitative results, such as area, power, and timing, and visual results, including the floorplan, placement, routing, and final layout.

The remainder of this report describes the RTL design, OpenLane implementation flow, physical implementation results, power and timing analysis, physical verification, and final GDSII layout.

2 Project Overview

The objective of this project was to understand and demonstrate a complete RTL-to-GDSII digital implementation flow using the open-source OpenLane framework. The design was developed in Verilog RTL and subsequently processed through synthesis, floorplanning, placement, clock-tree synthesis, routing, timing analysis, power estimation, and physical verification. The implementation was performed using the SKY130 process design kit and the `sky130_fd_sc_hd` standard-cell library [1, 3, 8].

The top-level design, `lowpower_soc`, integrates four main functional blocks that together form a small digital system-on-chip:

- a simple CPU for control and processing,
- a 256×8 -bit RAM for data storage,
- an 8-bit GPIO interface for external data output, and
- a timer for timing and control functions.

These functional blocks are integrated within the top-level `lowpower_soc` module and operate synchronously using a common clock and reset. The CPU provides the main control and processing function, while the RAM provides local data storage. The GPIO interface allows the internal system to produce 8-bit external output data, and the timer provides an additional timing-related function within the SoC.

The design also includes a sleep control mechanism intended to reduce unnecessary sequential activity when the system is inactive. The low-power control is implemented using the signal

```
assign system_enable = ~sleep;
```

When `sleep` is asserted, the system enable becomes inactive and the corresponding sequential logic stops its normal state updates. When sleep mode is released, the system resumes normal operation. Enable-based techniques are commonly used in low-power digital systems to reduce unnecessary switching activity during inactive periods [10].

The complete implementation was evaluated using the major physical-design metrics of area, power, and timing, together with DRC and LVS verification. The floorplan, placement, routing, and final GDSII layout were also inspected to understand how the RTL description is progressively transformed into a physical integrated-circuit implementation. The overall RTL architecture of the proposed low-power SoC is shown in Figure 1.

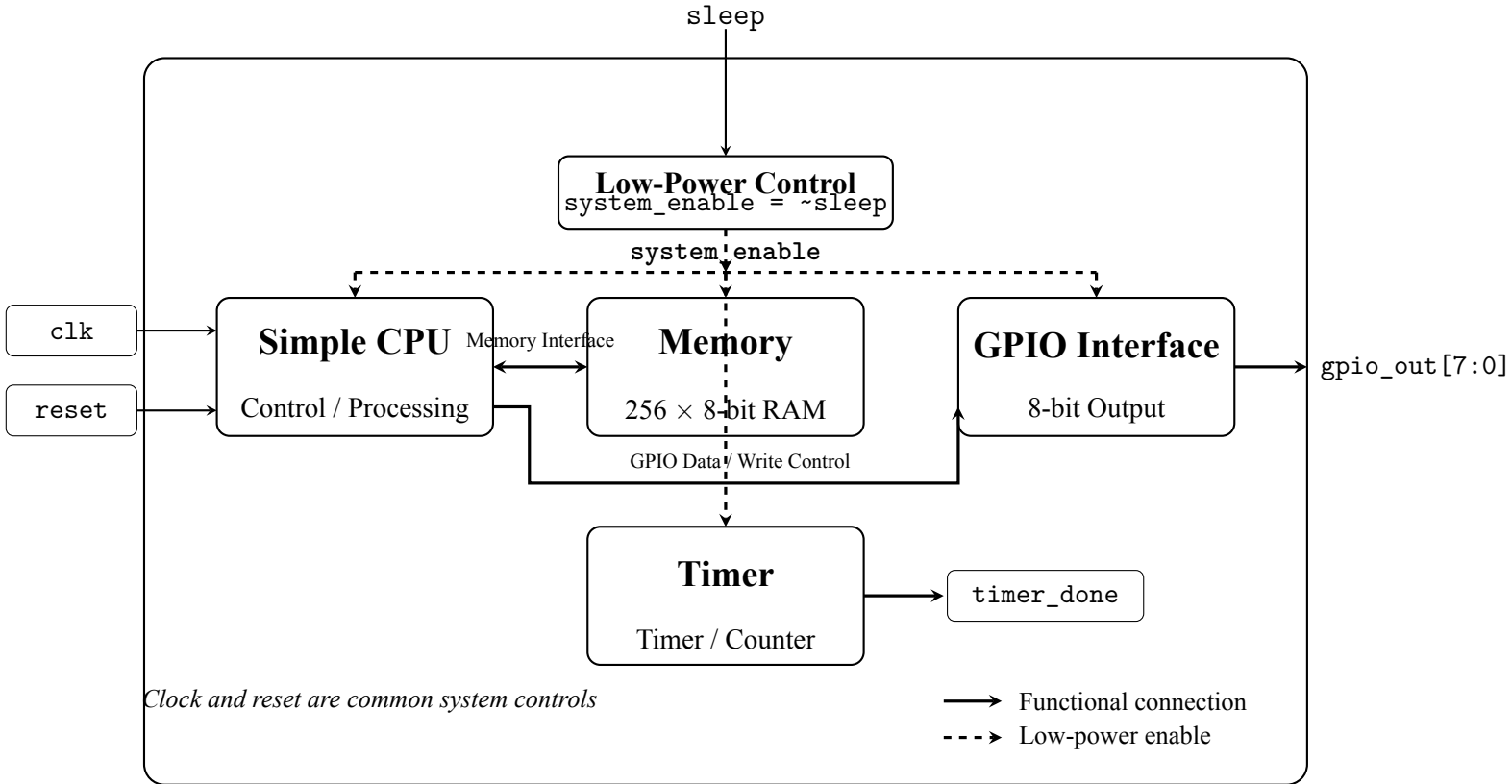


Figure 1: RTL architecture of the `lowpower_soc` design.

The architecture provides a compact integration of processing, memory, peripheral, timing, and low-power control functions within a single RTL design. The modular structure also simplifies the subsequent synthesis and physical implementation stages in OpenLane [1, 3].

3 RTL Design and Simulation

The RTL implementation of the `lowpower_soc` design was organized into separate Verilog modules, with each module performing a specific functional role. This modular structure simplifies the design, simulation, debugging, and subsequent synthesis process.

The RTL source files used in the project are:

- `lowpower_soc.v` – top-level SoC module,
- `simple_cpu.v` – CPU and control logic,
- `simple_ram.v` – 256×8 -bit memory,
- `gpio_interface.v` – 8-bit GPIO interface,
- `timer.v` – timer and counter logic, and
- `lowpower_soc_tb.v` – simulation testbench.

The top-level module integrates the CPU, memory, GPIO, and timer blocks and provides the external clock, reset, sleep, and output signals. The design operates synchronously, while the sleep input controls the system-enable mechanism.

The testbench was used to verify the functional behavior of the design by applying clock and reset signals and testing normal operation, sleep mode, wake-up, and reset behavior. Functional simulation was performed before synthesis to identify RTL-level issues prior to technology mapping and physical implementation [1, 3].

The RTL simulation waveform is shown in Figure 2.

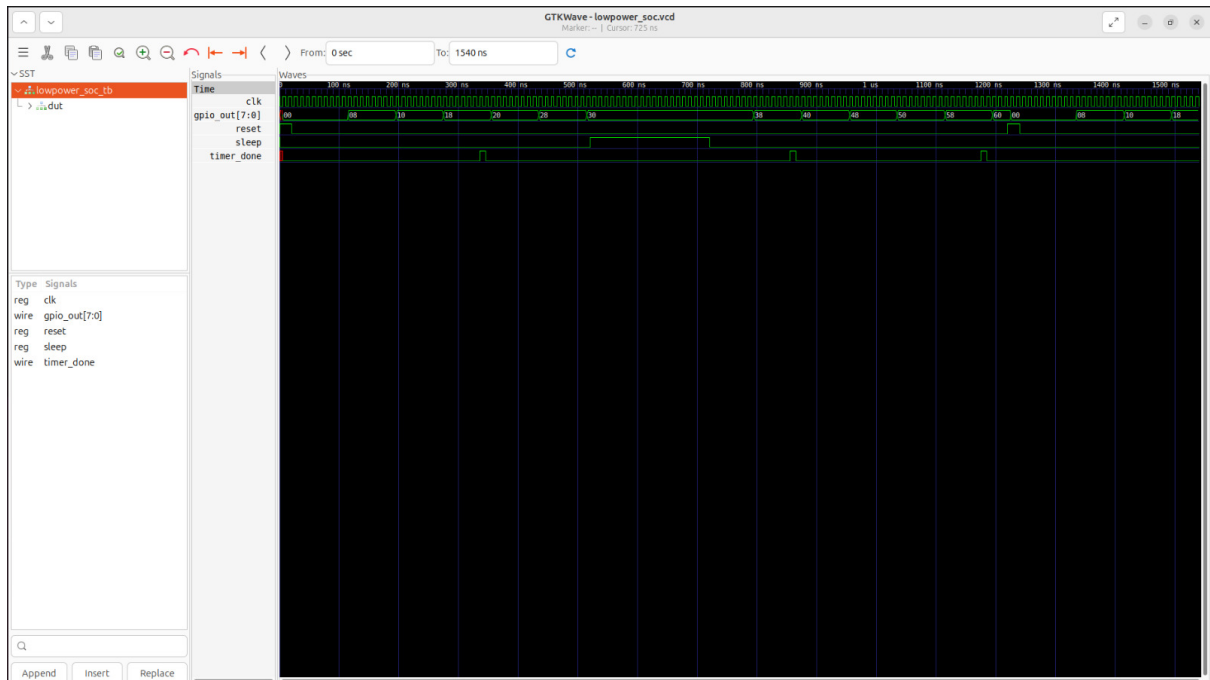


Figure 2: RTL simulation waveform of the `lowpower_soc` design.

The simulation stage therefore provided the functional baseline for the subsequent OpenLane implementation stages.

4 OpenLane RTL-to-GDSII Flow

After RTL simulation, the design was processed through the OpenLane physical-design flow. The major stages used in the implementation are shown in Figure 3.

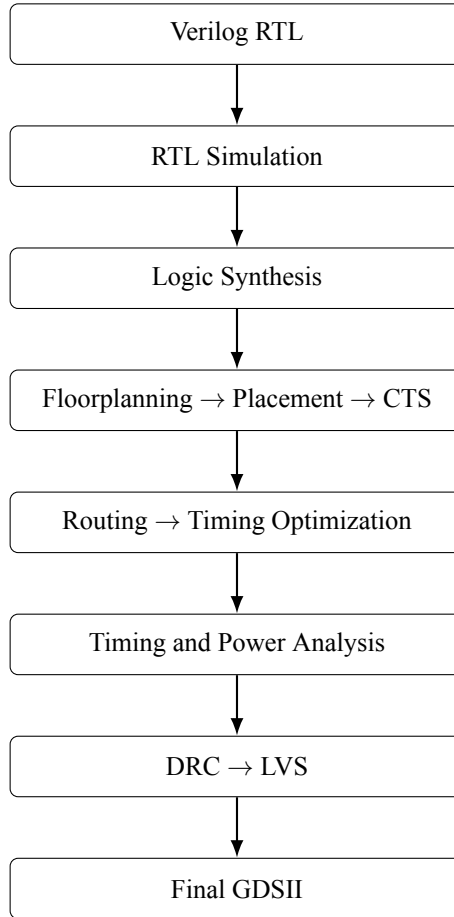


Figure 3: RTL-to-GDSII implementation flow used for the `lowpower_soc` design in OpenLane.

The synthesis stage converts the RTL into a technology-specific standard-cell netlist. Floorplanning defines the physical die and core regions, followed by standard-cell placement. Clock-tree synthesis constructs the clock distribution network, while timing optimization improves the implementation where necessary. Routing then establishes the physical interconnections between the cells.

After routing, static timing analysis and power estimation are performed to evaluate the implemented design. Finally, DRC and LVS are used to verify the physical layout before the final GDSII database is generated [1, 3, 5, 7].

The OpenLane flow therefore provides a continuous path from the original Verilog description to a physically implemented and verified layout, with intermediate reports available for evaluating each major stage of the implementation [1, 3].

The generated intermediate results were examined at each stage to understand the progression of the design from logical synthesis to its final physical representation and verification.

5 Physical Implementation Results

After synthesis, the design was taken through the main physical implementation stages of floorplanning, placement, clock-tree synthesis, and routing. These stages transform the synthesized standard-cell netlist into a physical representation while considering die dimensions, cell locations, clock distribution, interconnects, and technology-specific constraints [1, 3]. The generated physical views were examined to understand how the design evolved from a synthesized netlist to a routed layout.

The synthesis report shows that the design was mapped to 117 standard-cell instances with a synthesized cell area of $1350.0448 \mu\text{m}^2$. This represents the area of the technology-mapped standard-cell logic and provides the initial reference for evaluating the physical implementation. The synthesized cell count and area were obtained from the OpenLane synthesis report.

The initial floorplan report gives a die boundary of approximately

$$63 \mu\text{m} \times 73.72 \mu\text{m},$$

corresponding to a die area of approximately

$$4644.36 \mu\text{m}^2 = 0.00464 \text{ mm}^2.$$

The corresponding core area is approximately

$$2662.55 \mu\text{m}^2 = 0.00266 \text{ mm}^2.$$

The floorplanning stage defines the physical die and core regions within which the standard cells and routing resources are implemented. It provides the geometric foundation for placement and subsequent physical-design stages. The resulting floorplan is shown in Figure 4.

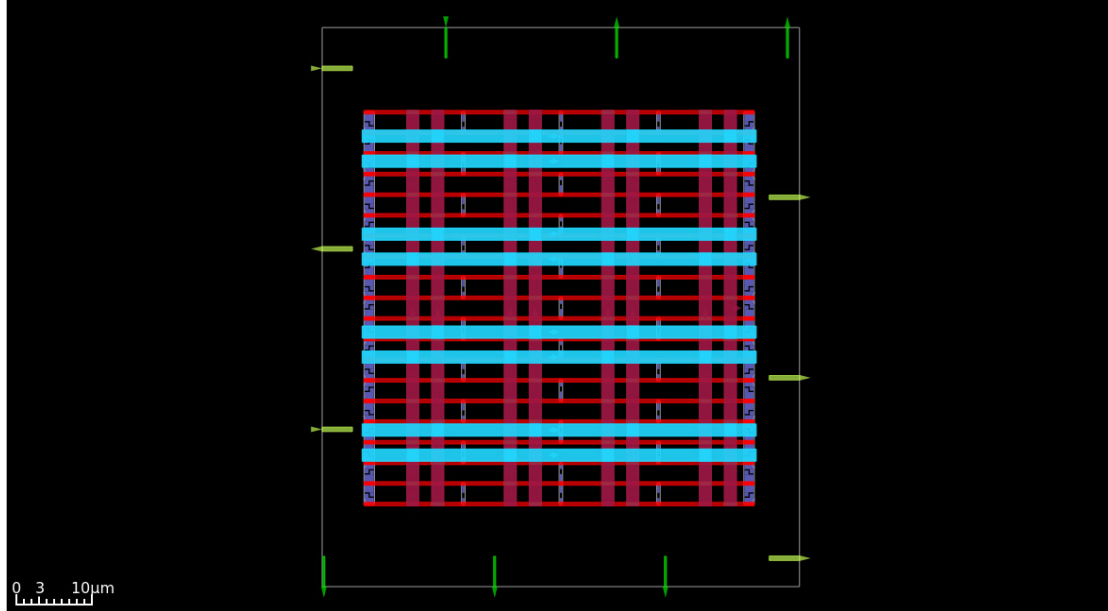


Figure 4: Initial floorplan of the lowpower_soc design, showing the defined die and core regions used for physical implementation.

Following floorplanning, the synthesized standard cells were placed within the available core region to

obtain a compact and routable arrangement [4, 3]. The resulting standard-cell placement is shown in Figure 5.

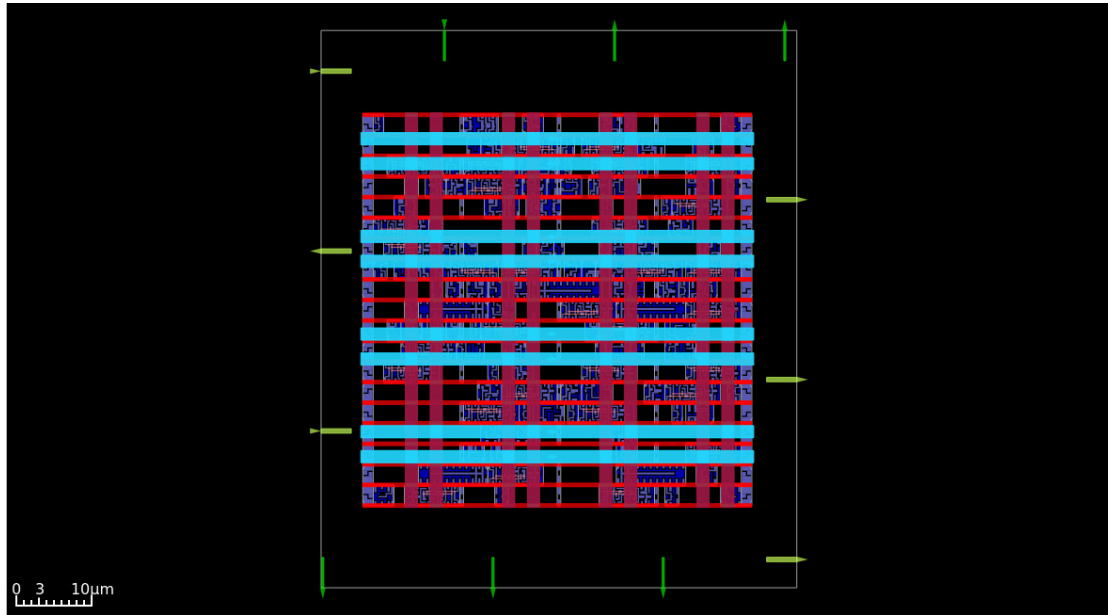


Figure 5: Standard-cell placement of the lowpower_soc design within the defined core region.

After placement and clock-tree synthesis, routing was performed to establish the physical interconnections between the standard cells while satisfying routing and technology constraints [5]. The final routed implementation is shown in Figure 6.

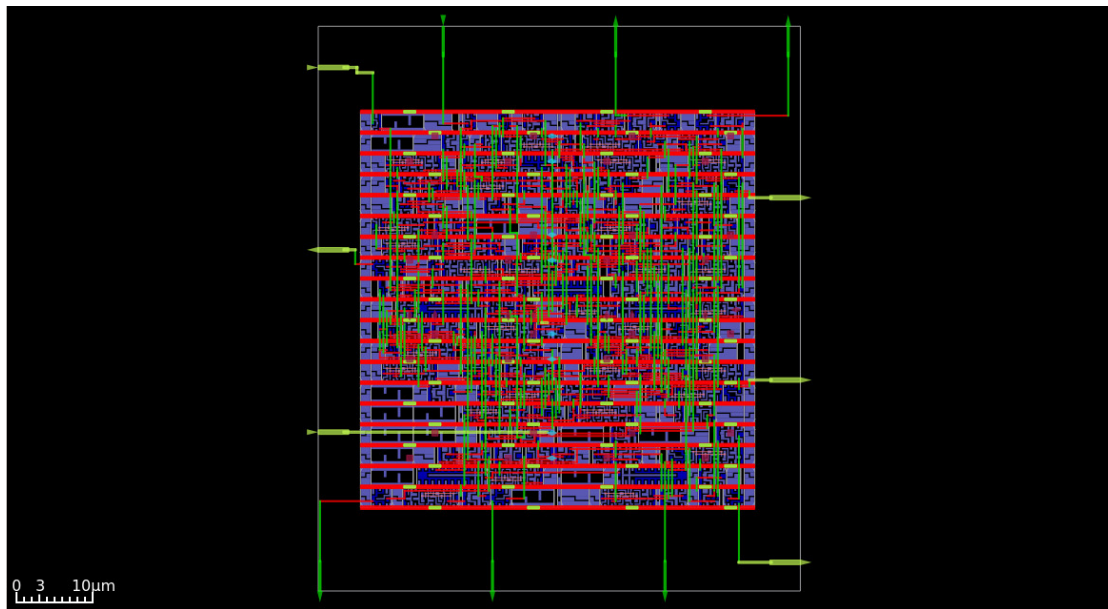


Figure 6: Final routed physical implementation of the lowpower_soc design.

The floorplan, placement, and routing views demonstrate the progression from the synthesized netlist to the physical implementation. These results were inspected before the final timing, power, DRC, LVS, and GDSII analysis.

6 Power, Timing and Verification Results

The major implementation results obtained from the OpenLane reports were analyzed in terms of area, power, timing, and physical verification. These metrics provide a quantitative assessment of the implemented design and complement the physical views presented in the previous section. Area indicates the physical size of the implemented logic, while timing and power provide information about the operating characteristics of the design. DRC and LVS provide additional evidence that the final physical implementation is consistent with the requirements of the target technology [1, 3, 7].

Overall Implementation Results

Table 1 summarizes the principal implementation results obtained from the synthesis and physical-design reports.

Table 1: Summary of area, timing, and physical implementation results.

Metric	Result
Synthesized standard cells	117
Synthesized cell area	1350.0448 μm^2
Core area	$\approx 2662.55 \mu\text{m}^2$
Die dimensions	63 $\mu\text{m} \times 73.72 \mu\text{m}$
Die area	$\approx 4644.36 \mu\text{m}^2$
Post-route setup slack	+6.41 ns
Post-route hold slack	+0.27 ns
Total negative slack	0
Worst negative slack	0
DRC violations	0
LVS errors	0

The synthesized design contains 117 standard-cell instances with a reported cell area of 1350.0448 μm^2 . This value represents the area of the technology-mapped standard-cell logic before considering the complete physical die dimensions. After floorplanning, the reported die dimensions were approximately 63 $\mu\text{m} \times 73.72 \mu\text{m}$, corresponding to a die area of approximately 4644.36 μm^2 .

The corresponding core area was approximately 2662.55 μm^2 . The difference between synthesized cell area, core area, and die area reflects the additional physical space required for placement, routing, power distribution, clock structures, and other implementation resources. The resulting physical dimensions therefore provide a more complete view of the implemented design than the synthesized cell area alone.

The final post-route timing results were also positive, with a setup slack of +6.41 ns and a hold slack of +0.27 ns. In addition, both total negative slack and worst negative slack were reported as zero. The combination of these results indicates that no reported setup or hold timing violations were present in the analyzed implementation.

Power Analysis

The OpenLane synthesis power report provides an estimated power breakdown at the typical corner. Power analysis is important for understanding how the implemented design consumes power and for distinguishing between internal cell power, switching power, and leakage power [3].

The total estimated power obtained from the report is approximately $152\ \mu\text{W}$. The complete power breakdown is presented in Table 2.

Table 2: Estimated power breakdown at the typical corner.

Power Component	Value
Internal power	$140\ \mu\text{W}$
Switching power	$11.9\ \mu\text{W}$
Leakage power	$0.509\ \text{nW}$
Total power	$152\ \mu\text{W}$

The internal power is approximately $140\ \mu\text{W}$, while the switching power is approximately $11.9\ \mu\text{W}$. The leakage power reported by the tool is approximately

$$5.09 \times 10^{-10}\ \text{W} = 0.509\ \text{nW}.$$

The internal component therefore represents the largest portion of the estimated power, followed by switching power. The leakage component is comparatively very small for this implementation and forms only a small fraction of the total reported power.

The power results are consistent with the use of a small standard-cell design containing a limited number of sequential and combinational elements. The sleep-controlled enable mechanism implemented at the RTL level is intended to prevent unnecessary state updates during inactive operation, which is consistent with the general objective of reducing unnecessary switching activity in low-power synchronous designs [10].

It is important to note that the values reported here are implementation-level estimates generated by the OpenLane power analysis at the typical corner. They should therefore be treated as estimated design power rather than measured power from fabricated silicon.

Timing Analysis

Static timing analysis was used throughout the physical implementation flow to monitor the setup and hold behavior of the design. Timing was evaluated at multiple stages so that the effect of placement, clock-tree synthesis, timing optimization, and routing could be observed [7, 3].

Table 3 summarizes the reported worst setup and hold slack values at the major physical-design stages.

Table 3: Timing results at different OpenLane implementation stages.

Implementation Stage	Setup Slack	Hold Slack
Global Placement (GPL)	+6.13 ns	+0.21 ns
Detailed Placement (DPL)	+6.18 ns	+0.20 ns
Clock Tree Synthesis (CTS)	+6.45 ns	+0.18 ns
Resizer / Timing Optimization	+6.31 ns	+0.31 ns
Global Routing (GRT)	+6.41 ns	+0.27 ns

The timing results remain positive throughout the reported implementation stages. After global placement, the design shows a setup slack of +6.13 ns and a hold slack of +0.21 ns. Following detailed placement, the setup and hold values remain positive at +6.18 ns and +0.20 ns, respectively.

After clock-tree synthesis, the setup slack increases to +6.45 ns, while the hold slack remains positive at +0.18 ns. The timing optimization stage then reports +6.31 ns setup slack and +0.31 ns hold slack. These results show that timing optimization maintained positive timing margins throughout the implementation process.

The final global-routing timing report gives a worst setup slack of +6.41 ns and a worst hold slack of +0.27 ns. Since both total negative slack and worst negative slack are zero, the analyzed implementation contains no reported setup or hold timing violations. The post-route result is therefore used as the main timing result for the completed physical implementation.

Physical Verification

After routing, physical verification was performed to check whether the final layout satisfied the relevant physical and technology constraints. Design-rule checking and layout-versus-schematic verification serve different purposes but together provide important evidence regarding the correctness of the physical implementation [9, 1].

DRC checks the physical geometry of the layout against the design rules of the target technology. The final DRC report for the design contained zero reported violations. This indicates that the analyzed layout did not contain the reported geometric or technology-rule violations.

LVS compares the connectivity represented by the extracted layout with the intended circuit netlist. The final LVS report reported zero errors and no net, device, pin, or property mismatches. This indicates that the extracted physical implementation was consistent with the intended circuit representation.

The final verification results are summarized in Table 4.

Table 4: Final physical verification results.

Verification Check	Result
Design Rule Checking (DRC)	0 violations
Layout Versus Schematic (LVS)	0 errors
Net mismatches	0
Device mismatches	0
Pin mismatches	0
Property mismatches	0

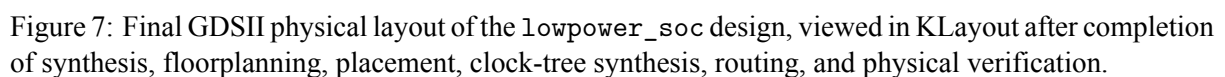
The zero-violation DRC result indicates that the final layout passed the reported design-rule checks, while the zero-error LVS result indicates that no connectivity or device mismatches were detected between the extracted layout and the intended design. Together, these results provide evidence that the final physical implementation is both geometrically valid and consistent with the implemented circuit structure.

Overall, the area, power, timing, and verification results show that the `lowpower_soc` design completed the OpenLane implementation flow successfully. The design achieved positive post-route setup and hold margins, an estimated total power of 152 μ W including 0.509 nW leakage power, and zero reported DRC violations and LVS errors. These quantitative results, together with the floorplan, placement, routing, and final GDSII views, provide a complete assessment of the implemented design.

The positive timing margins indicate that the implemented design maintains acceptable timing behavior after routing and optimization. The power results also provide a useful indication of the energy characteristics of the implemented standard-cell design. Overall, the reported results demonstrate the successful completion of the RTL-to-GDSII implementation and verification process using OpenLane.

The final stage of the OpenLane flow was the generation of the GDSII layout from the completed and physically verified design. GDSII is the physical layout representation containing the geometries of the implemented cells, metal interconnections, vias, power structures, and other layout features required to represent the fabricated circuit. The generated layout therefore represents the final physical realization of the `lowpower_soc` RTL design.

The final GDSII layout is shown in Figure 7.



The final layout provides a visual representation of the physical implementation obtained from the RTL design. The standard-cell instances can be observed within the core region, while the routed metal layers and vias establish the physical connectivity between the implemented circuit elements. The layout also shows the physical boundaries and power-distribution structures generated during the implementation process.

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8 Challenges and Key Learnings

This project provided practical experience with the complete digital implementation flow rather than focusing only on RTL development. One of the main learning outcomes was understanding how the different OpenLane stages transform a Verilog description into a technology-specific and physically implemented design [1, 3].

A major part of the learning process involved understanding the purpose of the intermediate reports generated during the flow. Synthesis reports were used to examine the mapped standard cells and area, while floorplan, placement, clock-tree, and routing results were used to understand the physical transformation of the design. Timing and power reports then provided quantitative information about the implemented circuit.

The project also demonstrated the relationship between RTL synthesis, physical planning, cell placement, clock-tree synthesis, routing, timing analysis, power estimation, and physical verification. Placement and routing decisions can affect interconnect length, congestion, and timing, making the physical implementation stages an important part of the overall design process [4, 5].

Another important learning outcome was understanding the difference between functional correctness and physical correctness. RTL simulation verifies the expected logical behavior of the design, whereas static timing analysis determines whether the implemented paths satisfy their timing constraints [7]. Similarly, DRC verifies geometric design rules, while LVS verifies the correspondence between the extracted layout and the intended circuit representation [9].

Overall, the work provided hands-on experience with an open-source RTL-to-GDSII flow and improved understanding of how RTL design decisions, physical implementation, timing, power, and verification are interconnected. The generated reports and physical views also provided a useful basis for analyzing each stage of the design process.

9 Conclusion

A complete RTL-to-GDSII implementation of the `lowpower_soc` design was successfully carried out using the OpenLane flow and the SKY130 technology [1, 8]. The project progressed from Verilog RTL and functional simulation through synthesis, floorplanning, placement, clock-tree synthesis, timing optimization, routing, power and timing analysis, DRC, LVS, and final GDSII generation.

The final implementation contained 117 synthesized standard-cell instances with a synthesized cell area of $1350.0448 \mu\text{m}^2$. The physical implementation produced a die measuring approximately $63 \mu\text{m} \times 73.72 \mu\text{m}$, corresponding to a die area of approximately $4644.36 \mu\text{m}^2$. These results demonstrate the physical size of the implemented SoC after synthesis and floorplanning.

The estimated total power at the typical corner was approximately $152 \mu\text{W}$, consisting of approximately $140 \mu\text{W}$ of internal power, $11.9 \mu\text{W}$ of switching power, and 0.509 nW of leakage power. The inclusion of a sleep-controlled enable mechanism provided a low-power feature at the RTL level, while the power analysis provided an implementation-level estimate of the resulting power characteristics.

Physical verification completed successfully, with zero reported DRC violations and zero LVS errors. The final GDSII layout was generated and inspected in KLayout, providing a visual representation of the completed physical implementation.

Overall, the project successfully demonstrated the transformation of a Verilog RTL design into a routed and physically verified GDSII layout using an open-source digital implementation flow. More importantly, it provided practical understanding of the relationship between RTL functionality, physical implementation, area, power, timing, and verification throughout the complete RTL-to-GDSII process [1, 3].

10 References

- [1] M. Shalan and T. Edwards, “Building OpenLANE: A 130nm OpenROAD-based Tapeout-Proven Flow,” in *Proceedings of the IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pp. 1–6, 2020. doi: 10.1145/3400302.3415735.
- [2] T. Ajayi, V. A. Chhabria, M. Fogaça, S. Hashemi, A. Hosny, A. B. Kahng, M. Kim, J. Lee, U. Mallappa, M. Neseem, G. Pradipta, S. Reda, M. Saligane, S. S. Sapatnekar, C. Sechen, M. Shalan, W. Swartz, L. Wang, Z. Wang, M. Woo, and B. Xu, “Toward an Open-Source Digital Flow: First Learnings from the OpenROAD Project,” in *Proceedings of the 56th Design Automation Conference (DAC)*, pp. 76:1–76:4, 2019. doi: 10.1145/3316781.3326334.
- [3] T. Ajayi, D. Blaauw, T.-B. Chan, C.-K. Cheng, V. A. Chhabria, D. K. Choo, M. Coltella, S. Dobre, R. Dreslinski, M. Fogaça, S. Hashemi, A. Hosny, A. B. Kahng, M. Kim, J. Li, Z. Liang, U. Mallappa, P. Penzes, G. Pradipta, S. Reda, A. Rovinski, K. Samadi, S. S. Sapatnekar, C. Sechen, V. Srinivas, W. Swartz, D. Sylvester, D. Urquhart, L. Wang, M. Woo, and B. Xu, “OpenROAD: Toward a Self-Driving, Open-Source Digital Layout Implementation Tool Chain,” in *Proceedings of the Government Microcircuit Applications and Critical Technology Conference (GOMACTech)*, pp. 1105–1110, 2019.
- [4] C.-K. Cheng, A. B. Kahng, I. Kang, and L. Wang, “RePlAce: Advancing Solution Quality and Routability Validation in Global Placement,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 38, no. 9, pp. 1717–1730, 2019. doi: 10.1109/TCAD.2018.2859220.
- [5] A. B. Kahng, L. Wang, and B. Xu, “TritonRoute: The Open Source Detailed Router,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2020. doi: 10.1109/TCAD.2020.3003234.
- [6] K. Han, A. B. Kahng, and J. Li, “Optimal Generalized H-Tree Topology and Buffering for High-Performance and Low-Power Clock Distribution,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2018. doi: 10.1109/TCAD.2018.2889756.
- [7] T.-W. Huang and M. D. F. Wong, “OpenTimer: A High-Performance Timing Analysis Tool,” in *Proceedings of the IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pp. 895–902, 2015. doi: 10.1109/ICCAD.2015.7372666.
- [8] Google and SkyWater, *SkyWater Open Source PDK: SKY130 Process Design Kit*, 2020. Available: <https://github.com/google/skywater-pdk>
- [9] J. K. Ousterhout, G. Hamachi, R. Mayo, and W. S. Scott, “Magic: A VLSI Layout System,” in *Proceedings of the 21st Design Automation Conference*, pp. 152–159, 1984. doi: 10.1109/DAC.1984.1585789.
- [10] D. Garrett, M. R. Stan, and A. A. Dean, “Challenges in Clockgating for a Low Power ASIC Methodology,” in *Proceedings of the International Symposium on Low Power Electronics and Design (ISLPED)*, 1999. doi: 10.1109/LPE.1999.145040.